

Module 8: Laboratory Activity 11

Title: Building a Simple Cloud-Based CRUD Web Application (PHP & MySQL)

Objective:

Students will create a basic **cloud-enabled CRUD (Create, Read, Update, Delete)** web application using PHP and MySQL, simulating a simple online data management system.

Materials Needed:

- XAMPP (Apache & MySQL)
- Text Editor (VS Code / Sublime)
- Web Browser (Chrome/Edge)

Procedure:

Part 1: Set Up Database

1. Open **phpMyAdmin**
2. Create a database named:
student_db
3. Create a table named students with fields:
 - id (INT, AUTO_INCREMENT, PRIMARY KEY)
 - name (VARCHAR 100)
 - course (VARCHAR 100)

Part 2: Create Database Connection

Create config.php:

```
<?php
$conn = new mysqli("localhost", "root", "", "student_db");
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

Part 3: Create Form (Create Function)

Create index.php:

```
<?php include 'config.php'; ?>
```

```
<form method="POST">
    Name: <input type="text" name="name" required><br>
    Course: <input type="text" name="course" required><br>
    <button type="submit" name="submit">Add</button>
</form>
```

```
<?php
if (isset($_POST['submit'])) {
    $name = $_POST['name'];
    $course = $_POST['course'];

    $conn->query("INSERT INTO students (name, course) VALUES ('$name', '$course')");
}
```

?>

Part 4: Display Records (Read Function)

Add below the form:

```
$result = $conn->query("SELECT * FROM students");
```

```
while ($row = $result->fetch_assoc()) {  
    echo $row['id'] . " - " . $row['name'] . " - " . $row['course'] . "<br>";  
}
```

Part 5: Delete Function

Add delete link:

```
echo "<a href='delete.php?id=".$row['id']."'>Delete</a><br>";
```

Create delete.php:

```
<?php
```

```
include 'config.php';
```

```
$id = $_GET['id'];
```

```
$conn->query("DELETE FROM students WHERE id=$id");
```

```
header("Location: index.php");
```

```
?>
```

Part 6: Run the System

1. Start Apache and MySQL in XAMPP
2. Open browser:
http://localhost/your-folder-name/index.php
3. Add, view, and delete student records

Expected Output:

- Working web form
- Data stored in MySQL database
- Display of records
- Delete functionality working

Extension Task

- Add **Update/Edit function**
- Apply **Bootstrap styling**
- Validate inputs using JavaScript
- Deploy to a free cloud hosting service (e.g., 000webhost)